

Search Bar Test Plan

Phase 1 — Requirement Analysis

Prior to designing test boundaries, key clarifying questions must be posed to the Product Owner to baseline expected behavior:

1. **Character Constraints:** What is the maximum character limit and input validation logic for the search bar?
 2. **Text Wrap & Layout:** How should the system handle extremely long product titles in the results view?
 3. **Rich Media Inputs:** Can an image be pasted into the search bar (Visual Search support)?
 4. **Fallback Scenarios:** What is the expected behavior when a search yields zero results?
 5. **Data Loading:** What are the pagination thresholds and lazy-loading rules for the results list?
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Phase 2 — Test Planning

Scope Boundaries

- **In-Scope:** Verification of the search bar input field functionality. This includes text entry limits, input validation (alphanumeric and special characters), copy/paste behavior (including handling of non-text items like images), auto-suggestion dropdown behavior, and the trigger mechanism for submitting a search query.
- **Out-of-Scope:** The search results display page, including product cards, filtering panels, pagination layout, sorting logic, and mixed-category result prioritization.

Project Risk & Mitigation

- **Risk:** *Unclear handling of multimedia clipboard data.* If users attempt to paste an image directly into the search input and the system lacks a specific rejection or handling mechanism, it could cause browser performance lag, unexpected UI freezes, or unhandled 500 server errors.
 - **Mitigation Strategy:** Request explicit confirmation from the Product Owner regarding visual search support prior to test case execution. If unsupported, prioritize boundary testing around clipboard actions to ensure the system fails gracefully with an appropriate error message or rejects the paste outright.
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Phase 3 — Test Case Design

The core test execution matrix focuses on critical boundary conditions and input sanitization:

- **TC_Search_001:** Validate maximum character length boundaries for the search input.
- **TC_Search_002:** Validate system behavior and error handling when rich media (images) are pasted into the search bar.
- **TC_Search_003:** Validate input validation constraints for special characters and emojis.

Phase 4 — Environment Setup

To ensure consistent cross-platform functionality and responsiveness of the search input field, testing will be executed across the following hardware and software configurations:

Desktop Testing Matrix	Mobile & Tablet Testing Matrix (Responsive UI)
Operating Systems: macOS (Latest), Windows 11	iOS Ecosystem: iPhone (Latest iOS / Safari), iPad (Latest iPadOS / Safari & Chrome)
Target Browsers: Google Chrome, Safari, Microsoft Edge	Android Ecosystem: Android Mobile Phone (Latest OS / Google Chrome)

Phase 5 — Execution

Execution Timeline

- **Start Date:** 06/06/2026
- **End Date:** 06/30/2026

Entry Criteria

Testing will only officially commence once the following quality gates are successfully met:

1. **Code Deployment:** The Search Bar feature branch must be fully compiled and successfully deployed to the dedicated QA Staging environment.
 2. **Requirements Baseline:** The Product Owner must formally sign off on the minimum/maximum character limits and visual search clipboard requirements.
 3. **Sanity Check:** A quick smoke test must pass, confirming that the Staging environment is up, reachable, and the search input box renders correctly without crashing the browser.
 4. **Test Documentation:** All 3 core test cases from Phase 3 must be reviewed and approved for execution.
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Phase 6 — Closure

The final phase summarizes execution results and determines release readiness using a clean closure framework:

- **Execution Overview:** A high-level summary of the total number of test cases executed (e.g., 3 planned), the final pass/fail ratio, and a list of any deferred or blocked test cases.
- **Defect Metrics & Status:** A breakdown of all logged defects discovered during the search input testing, categorized by severity (Critical, High, Medium, Low), along with their current resolution status (e.g., Fixed, Retested, Open).
- **Sign-Off & Recommendation:** A definitive statement confirming whether the Search Bar feature has met the quality gates and is recommended for deployment to Production, or if it should be held back due to outstanding critical risks.